WORKED SOLUTION

STEP I 2008, Question 12

Probability and Statistics • Maximum of Discrete Uniforms

STEP I 2008, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s1-q12

The key idea

X is just the larger of the two scores. Asking for "the larger equals r" directly means counting several cases, but "the larger is *at most* r" is easy: it happens exactly when *both* scores are at most r. So we work with the cumulative distribution $\mathrm{P}(X \leq r)$ first and take differences. The same cumulative distribution then makes the median almost immediate.

Solution

Let X_1 and X_2 be the two independent scores, and $X = \max(X_1, X_2)$ (the two cases in the definition agree when $X_1 = X_2$, so X is well defined).

Part (i)

For $r = 0, 1, \ldots, N$, the event $\{X \leq r\}$ is the event $\{X_1 \leq r \text{ and } X_2 \leq r\}$. Each of these has probability r/N, and they are independent because X_1 and X_2 are, so

$$\mathrm{P}(X \leq r) = \frac{r}{N} \cdot \frac{r}{N} = \frac{r^2}{N^2}.$$

This holds for $r = 0$ too, giving 0. Hence, for $r = 1, 2, \ldots, N$,

$$\mathrm{P}(X = r) = \mathrm{P}(X \leq r) - \mathrm{P}(X \leq r-1) = \frac{r^2 - (r-1)^2}{N^2} = \frac{2r-1}{N^2},$$

as required. (As a sanity check, $\sum_{r=1}^{N}(2r-1) = N^2$, so these add to 1.)

Part (ii)

Using the two given sums,

$$\begin{aligned}\mu = \sum_{r=1}^{N} r\,\frac{2r-1}{N^2} &= \frac{1}{N^2}\left(2\sum_{r=1}^{N} r^2 - \sum_{r=1}^{N} r\right)\\ &= \frac{1}{N^2}\left(\frac{N(N+1)(2N+1)}{3} - \frac{N(N+1)}{2}\right)\\ &= \frac{N+1}{N}\cdot\frac{2(2N+1)-3}{6} = \frac{(N+1)(4N-1)}{6N}.\end{aligned}$$

When $N = 100$,

$$\mu = \frac{101\times 399}{600} = \frac{40\,299}{600} = 67.165,$$

as required.

Part (iii)

From part (i), $\mathrm{P}(X \le m) = m^2/N^2$ and $\mathrm{P}(X \ge m) = 1 - \mathrm{P}(X \le m-1) = 1 - (m-1)^2/N^2$. The two conditions on the median become

$$\mathrm{P}(X \le m) \ge \tfrac{1}{2} \iff m^2 \ge \tfrac{1}{2}N^2 \iff m \ge \frac{N}{\sqrt{2}},$$

$$\mathrm{P}(X \ge m) \ge \tfrac{1}{2} \iff (m-1)^2 \le \tfrac{1}{2}N^2 \iff m-1 \le \frac{N}{\sqrt{2}},$$

taking positive square roots, which is valid since $m \ge 1$ and so m and $m-1$ are non-negative. So m must be an integer with

$$\frac{N}{\sqrt{2}} \le m \le \frac{N}{\sqrt{2}} + 1.$$

Ask yourself An interval of length 1 can contain two integers. Could there be two medians?

Only if both ends were integers, which would need $N/\sqrt{2}$ to be an integer. But $N/\sqrt{2} = k$ would give $\sqrt{2} = N/k$, a rational number, which is impossible. So both inequalities are strict, the open interval of length 1 contains exactly one integer, and

$$m = \left\lceil \frac{N}{\sqrt{2}} \right\rceil,$$

the smallest integer greater than $N/\sqrt{2}$.

When $N = 100$, $N/\sqrt{2} = 50\sqrt{2} \approx 70.71$, so $m = 71$. Directly: $71^2 = 5041 \ge 5000 = \frac{1}{2}\times 100^2$ and $70^2 = 4900 \le 5000$, so both conditions hold.

Part (iv)

From part (iii), $m = N/\sqrt{2} + \theta$ for some $0 < \theta < 1$, so

$$\frac{m}{N} = \frac{1}{\sqrt{2}} + \frac{\theta}{N} \to \frac{1}{\sqrt{2}} \quad \text{as } N \to \infty.$$

From part (ii),

$$\frac{\mu}{N} = \frac{(N+1)(4N-1)}{6N^2} = \frac{1}{6}\left(1 + \frac{1}{N}\right)\left(4 - \frac{1}{N}\right) \to \frac{4}{6} = \frac{2}{3}.$$

Therefore, for large N,

$$\frac{\mu}{m} = \frac{\mu/N}{m/N} \approx \frac{2/3}{1/\sqrt{2}} = \frac{2\sqrt{2}}{3}.$$

Checks

- With $N = 2$: the four equally likely pairs give $X = 1$ once and $X = 2$ three times, matching $\frac{1}{4}$ and $\frac{3}{4}$ from part (i), and $\mu = \frac{7}{4}$, matching $\frac{3\times 7}{12}$.
- Listing all N^2 pairs by computer for $N \leq 40$ confirms parts (i) and (ii) exactly; testing every integer against the median definition for $N \leq 2000$ always finds exactly one median, equal to $\lceil N/\sqrt{2} \rceil$.
- For $N = 10^5$, $\mu/m \approx 0.942812$, against $2\sqrt{2}/3 \approx 0.942809$.

Common slips

- Counting $\mathrm{P}(X = r)$ as $2 \times \frac{r}{N} \cdot \frac{1}{N}$, forgetting that the case $X_1 = X_2 = r$ is then counted twice; the correct count is $2(r-1) + 1 = 2r - 1$.
- Writing $\mathrm{P}(X \geq m) = 1 - m^2/N^2$: that is $\mathrm{P}(X > m)$, not $\mathrm{P}(X \geq m)$.
- Giving $m = 70$ for $N = 100$ by rounding 70.71 to the nearest integer rather than up.

Beyond STEP

X is the simplest example of an *order statistic*: the maximum of a sample. As $N \to \infty$, X/N approaches a continuous random variable on $[0, 1]$ with density $2x$, whose mean is $\frac{2}{3}$ and whose median is $1/\sqrt{2}$, which is exactly why the ratio in part (iv) appears. Order statistics of larger samples lead to the beta distributions. Blitzstein and Hwang, *Introduction to Probability*, treats order statistics in an accessible way.

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